

CampaignX: AI Multi-Agent System for Marketing Campaign Automation

InXiteOut | IIT Mandi Tech Fest

This document acts as the Rule Book for **CampaignX** hackathon being conducted by InXiteOut as part of FrostHack, XPECTO 2026 Tech Fest at IIT Mandi. The rules laid out in this document governs the terms and conditions of this hackathon event.

1 Objective

Build AI Agent-based web application for end-to-end Digital Marketing Campaign Management for **SuperBFSI**, an Indian BFSI service provider catering to customers located across India. The application should have human-in-the-loop for critical approvals and a functional UI.

2 Prizes

2.1 Monetary Rewards for the Winning Teams

- 1st Prize: INR 25,000 to the respective team
- 2nd Prize: INR 15,000 to the respective team
- 3rd Prize: INR 10,000 to the respective team
- Vouchers worth INR 30,000 in total to other good teams

For matters regarding the selection of winners, InXiteOut's decision will be final and binding.

2.2 Other Potential Rewards to Outstanding Individuals

The submissions will be judged by the co-founder(s) of InXiteOut. If you truly stand out, we may explore the following possible benefits **based on merit and alignment**:

- **Paid Internship opportunity** (for 3-6 months duration) in our AI Engineering dept.
- Opportunity to appear directly in the **Final Technical Round of our Off-Campus Interview** (i.e., by skipping the written test and initial technical rounds) for Full-Time Hiring in our AI Engineering dept (this is applicable for only final year students who, if selected, can join us within 6 months of the offer date).

For matters regarding Internships and Off-Campus Interviews, InXiteOut's decision will be final and binding.

3 Background

Modern marketing campaigns require coordination across multiple channels: email, text message, social media, paid ads, landing pages, and analytics platforms. Campaign managers often spend more time on repetitive tasks such as data gathering, segmentation, scheduling, and reporting than on optimization strategy and creativity.

AI agents have the potential to transform this process by acting as autonomous campaign assistants that can plan, execute, monitor, and optimize campaigns. However, building such agents requires integrating multiple APIs, handling real-time feedback, and making intelligent decisions based on data.

This hackathon challenges participants to design and build an AI-based multi-agent system for managing digital (**Email only**) marketing campaigns for **SuperBFSI**, a BFSI service provider catering to customers located across India. The solution needs to be delivered as a web application with a functional UI that allows human-in-the-loop approach for critical approvals. The scope is limited to India only.

4 Core Problem Statement

Build an AI agent solution that can **plan, launch, monitor, and optimize a digital marketing campaign (via email), for launching a new Term Deposit product for SuperBFSI**. The system should be capable of **automatically**:

- i. Understanding the campaign brief as provided by the marketer as natural language text input via the UI (e.g., *“Run email campaign for launching XDeposit, a flagship term deposit product from SuperBFSI, that gives 1 percentage point higher returns than its competitors. Announce an additional 0.25 percentage point higher returns for female senior citizens. Optimise for open rate and click rate. Don’t skip emails to customers marked ‘inactive’. Include the call to action: <https://superbfsi.com/xdeposit/explore/>.”*).
 - a. **Note:** The URL mentioned above is dummy.
- ii. Identifying campaign strategy for optimal performance metrics (open rate and click rate).
 - a. **Note:** Campaign performance metrics (i.e., open rate and click rate) may depend on various factors such as customer demography, behavioural segmentation, time slot when the email campaign was sent, email content (including but not limited to style, tone, font, length etc.).
- iii. Generating campaign assets/contents/creatives according to the campaign strategy identified.
- iv. Requesting UI-based human-in-loop approval for executing the strategy identified.
- v. Integrating and leveraging campaign management and reporting APIs as described in **Section 5.2** to:
 - a. Execute campaign actions.
 - b. Collect performance metrics (open rate and click rate).
- vi. Analysing performance metrics.
- vii. Iterating steps ii-vi.

5 Tech Stack

- Use programming language(s), agentic framework(s), libraries, tools of your choice.
- Cloud deployment is not mandatory. But a live demonstration is needed (from a local system if cloud deployment is not done).
- Free tier / trial access of the following tools (a non-exhaustive, tentative list) can be used (however, participants may as well use other tools of their choice):
 - Cloud deployment (optional)
 - Appwrite: <https://appwrite.io/pricing>
 - Render: <https://render.com/pricing>
 - Netlify: <https://www.netlify.com/pricing/>
 - Vercel: <https://vercel.com/pricing>
 - Databases (if/as required)
 - MongoDB
 - SQLite / MySQL
 - PostgreSQL

5.1 GenAI Tools

Use open-source LLMs and/or free tier / trial LLM APIs of any provider (e.g., Gemini / Groq / Mistral / OpenRouter / Cohere / AnyScale / HuggingFace / AIMLAPI etc.).

An indicative list (non-exhaustive) along with best practices and key setup guides is captured here: [LLM API Key Setup Guide](#).

However, participants are free to use APIs / LLMs of their choice.

5.2 Email Campaign Management APIs

API Documentation:

- [Downloadable OpenAPI Specification in ReDoc format](#).
- [PDF document](#).

These APIs are made available to the participants for the purposes of this hackathon.

A daily rate limit of 100 calls per day per registered team for each endpoint apply. Only one signup should be done per team. For more information, please refer to the API documentation link mentioned above.

5.3 Tentative Solution Architecture

The solution may have at least the following components:

- Backend that supports agentic automation of:
 - Customer profiling
 - Campaign strategy for optimal performance metrics (open rate and click rate)
 - Campaign content generation
 - Campaign sending/scheduling (subject to human-in-loop approval)
 - Campaign report data collection
 - Campaign report data analysis
- Frontend involving:

- Simple, functional UI that allows human-in-loop decision-making before sending/scheduling campaigns

6 Required Functional Capabilities

6.1 Customer Cohort

Get customer cohort by using APIs as described in **Section 5.2**. This will act as the customer master data including customer ids and their demographic details.

6.2 Campaign Brief

The agent system must be able to parse a campaign brief to be provided as an input (**as free-flowing natural language text**) by the marketer to the application via UI (e.g., “Run email campaign for launching XDeposit, a flagship term deposit product from SuperBFSI, that gives 1 percentage point higher returns than its competitors. Announce an additional 0.25 percentage point higher returns for female senior citizens. Optimise for open rate and click rate. Don’t skip emails to customers marked ‘inactive’. Include the call to action: <https://superbfsi.com/xdeposit/explore/>.”)

Note: The URL mentioned above is dummy.

6.3 Campaign Planning

The agent system must be able to generate **automatically**:

- Campaign strategy
 - **Note:** To allow the agent system to make automated optimizations based on results in order to enhance the campaign performance metrics, it is recommended to use A/B testing or similar methodologies.
- Customer segment (i.e. list of customer IDs) to which the campaign should be sent
- Time to send the campaign

6.4 Content Generation

The email campaign body may contain only the following types of elements:

- Any text in English
- Any emoji
- A specific URL: <https://superbfsi.com/xdeposit/explore/>

The email campaign subject header may contain only the following types of elements:

- Any text in English

The agent system should **automatically**:

- Generate text variations (title and body content, style, tone, length etc.)
- Decide if any font variations (viz., underline, bold, italicize) to be done in the content and if yes, what and where in the content; and do accordingly
- Decide whether to include any emoji and if yes, what and where in the content; and do accordingly

- Decide whether to include the URL mentioned above and if yes, where in the content; and do accordingly

6.5 Human-in-Loop Approval

The application must:

- Display campaign details (including content, list of customers to whom this content will be sent, time to send etc.) on UI to enable informed decision-making by human
- Allow approval / rejection

6.6 Campaign Scheduling

The agent system must leverage APIs as described in **Section 5.2** to schedule the campaign on any future date & time.

Note:

- For the purposes of this hackathon, scheduling a campaign will be considered as equivalent to executing the campaign.
- **Deterministic API calling should be avoided.** Rather, for true agentic workflow, API documentation-based automatic and dynamic discovery for tool calling must be implemented.

6.7 Performance Monitoring

The agent system must **automatically**:

- Fetch campaign performance report (opens and clicks) through APIs as described in **Section 5.2**.
- Analyse them.

Note: For the purposes of this hackathon, it will be possible to fetch campaign performance report immediately after the campaign is scheduled (even if the scheduled time is still in future). This is to expedite the development process by allowing participants to receive gamified performance metrics (in alignment with various campaign factors) immediately without needing to wait.

6.8 Autonomous Optimization

Based on performance metrics (open rate and click rate), the agent system should **automatically**:

- Identify micro-segments
 - Generate optimized variants
 - Adjust:
 - Email (content, style, tone, font etc.)
 - Send time (scheduling on a future time is possible)
- Relaunch:
 - Send updated / new campaign to specific segment(s) or entire cohort (subject to human-in-loop approval)

7 Submission

7.1 Phases

We are dividing the hackathon timeline into two distinct phases:

- Development phase (2 weeks long): From the start of CampaignX Rule Book launch (on 27th February 2026) till 13th March 11:59:59 PM.
- Test phase (2 days long): 14th March 12:00:00 AM to 16th March 2026.
 - **Note:** The customer cohort returned by the API during the Test phase might be different from the customer cohort returned by the API during the Development phase. That is, the **customer cohort might be changed on 14th March 12:00:00 AM. Participants must retrieve the new customer cohort for all the deliverables described below and avoid using the old customer cohort.**

7.2 Deliverables

- **Code** in any Git-based repository for the web application prototype.
 - **Note:** A basic MVP solution is all that's required. Scalability is out of scope.
- **Screen recording session (video less than 3 minutes long)** showcasing the end-to-end working of the application. Teams **must** demonstrate at least one automated campaign optimization loop during the screen recording session. Teams **must** showcase code snippets that highlight API documentation-based dynamic discovery for campaign management API tool calling.
 - **Note:** The recording must happen during the Test phase.
- **Slide deck (not more than 5 slides excluding the cover slide and thank you slide)**
 - An executive summary of customer insights based on campaigns runs during the Test phase (**not more than 5 bullets**).
 - Architecture, workflow, and tech stack overview.

7.3 Submission Mechanism, Format, and Deadline

Participants will get **2 weeks** of time to develop their solution.

The deliverables mentioned in **Section 7.2** will need to be submitted by **14th March 2026, 11:59:59 PM**, by sending an email to campaignx@inxiteout.ai, with only **<team name>** mentioned as the subject header.

The **email body should include nothing but 3 links**:

- i. Publicly accessible link to code repository (e.g., GitHub) containing the code developed
- ii. Publicly accessible link to screen recording video hosted on 3rd party storage location (e.g., YouTube, Google Drive etc.)
- iii. Publicly accessible link to deck hosted on 3rd party storage location (e.g., Google Drive).

Notes:

- All the links to be mentioned in the email body must be publicly accessible. Otherwise, the team will be automatically disqualified. Teams are requested to take caution to ensure and validate that the links included in the email are correct and publicly accessible.

- Only one submission per team is allowed. In case of duplicate submissions by the same team, only the latest submission will be considered.
- Submissions after the deadline (i.e., after 14th March, 11:59:59 PM) won't be accepted.

8 Evaluation

The evaluation will happen in two steps.

8.1 Step 1: Shortlisting

8.1.1 Dates

Based on the Test phase Submission as described in **Section 7.3**, **top ten (10) teams will be shortlisted** to make presentations in front of InXiteOut judges at IIT Mandi campus. The names of the shortlisted teams will be announced by **15th March 2026 11:59:59 PM**.

Any deviations from the rules and guidelines laid out in this document will result in disqualification.

8.1.2 Criteria

Criterion	Weight
Campaign performance metrics	50%
Functionality & completeness	30%
Deliverable quality	20%

Notes:

- The open rate and click rate of the entire relevant customer cohort, based on the latest campaigns executed during the Test phase, will be considered as campaign performance metrics.
 - That is, **the minimum set of latest campaigns that covered the entire customer cohort during the Test phase will be considered.**
- The **weightage distribution** across open rate and click rate while computing the scores for campaign performance metrics criteria will be:
 - Click rate: 70%
 - Open rate: 30%

8.1.3 Red Flags

- Screen recording with the old customer cohort that existed during Development phase (i.e., before 14th March 12:00:00 AM).
- Only a chatbot or copy generator.
- No agentic integration of campaign management APIs (i.e., deterministic API calling instead of API documentation-based dynamic discovery for tool calling).
- Manual optimization instead of decisions by agent system.
- Hardcoded campaign flows without automated reasoning.
- No human-in-loop.

8.2 Step 2: Live Presentation

8.2.1 Date

The shortlisted teams will make presentations in front of InXiteOut judges at IIT Mandi campus on **16th March 2026**.

8.2.2 Format

- Each team will get a maximum of 15 mins in total:
 - Up to 6 mins for **Presentation**
 - Followed by up to 4 mins for a **Live Demo**
 - **Note:** The campaign brief (see **Section 6.2**) may be requested to be changed by the judges during the Live Demo on an ad-hoc basis to validate the agentic workflow quality and stability.
 - Followed by 5 mins for **Q&A**
- The Presentation should:
 - Explain the architecture and agentic workflow
 - Highlight the differentiating factors (along with appropriate substantiations) of the application developed by the team

8.2.3 Criteria

Criterion	Weight
Campaign performance metrics	20%
Quality of AI agent system logic	20%
User experience & stable execution	15%
Code modularity and readability	15%
Presentation skills	15%
Innovation & creativity	15%

Note:

- For the campaign performance metrics criteria, the scores from the Shortlisting step will be used as is.
- However, the remaining criteria will be scored based on Live Presentation performance.

8.2.4 Bonus Points (Optional)

- Logging of agent interaction / decision / reasoning data including input and output.
- Real-time dashboards of campaign performance metrics.
- Cloud deployment.

Note: Achieve excellence in criteria mentioned in **Section 8.2.3** before focusing on Bonus Points.

8.3 Result Announcement

The final result will be announced on **16th March 2026** based on Live Presentation performance.